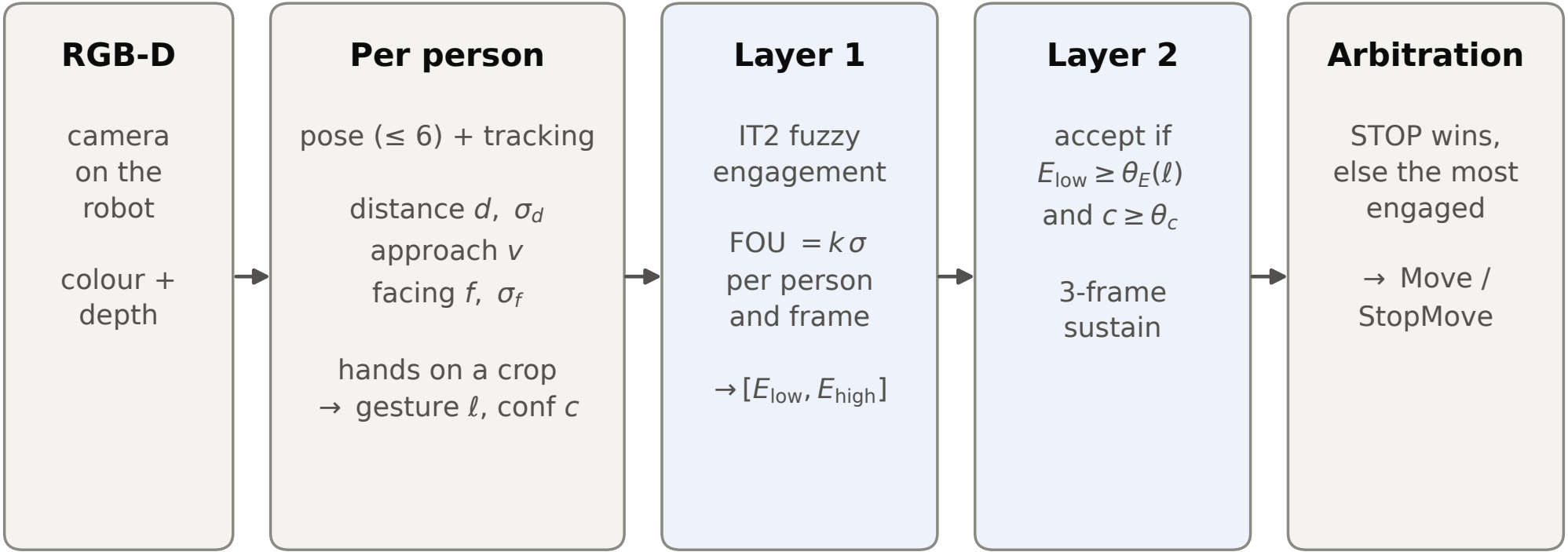
RGB-D
camera on the robot
colour + depth
Per person
pose (≤ 6) + tracking
distance d, σ_d
approach v
facing f, σ_f
hands on a crop
→ gesture ℓ, conf c
Layer 1
IT2 fuzzy engagement
FOU $= k\,\sigma$
per person and frame
$\rightarrow [E_{\text{low}}, E_{\text{high}}]$
Layer 2
accept if
$E_{\text{low}} \geq \theta_E(\ell)$
and $c \geq \theta_c$
3-frame sustain
Arbitration
STOP wins, else the most engaged
→ Move / StopMove